A Discrete Chern–Weil Proposal:

Periodic Cyclic Cohomology of the Universal Nest Algebra, and a Čech Comparison Theorem

Working proposal — discussion draft, v2

July 2026

Abstract

This note is not a construction; it is a plan for one, written to expose its open joints. The object of interest is a *relational substrate*: a nest of local operator algebras glued by strict comparison isomorphisms over an abstract cover. The goal is a discrete Chern–Weil theory on this substrate — a mechanism producing characteristic classes from purely relational, non-local data — modelled on the local Riemann–Roch–Hirzebruch (LRRH) architecture of Willwacher [1] and its antecedents [2, 3, 4, 5]. Relative to the first draft of this proposal, the receptacle has changed, and the change simplifies everything downstream. The primitive object is now the *universal nest algebra* $\mathbb{A}_u$ (§3), in which every comparison of the substrate is an inner symbol and flatness is a defining relation; the intrinsic receptacle is its periodic cyclic theory $\mathrm{HP}(\mathbb{A}_u)$, carried by the *single* Connes differential $b + B$ [6, 10]. There is no mixed coboundary and no bicomplex in the foundations. The Čech–NCG bicomplex of the first draft survives as a *comparison model*: filtering cyclic chains of $\mathbb{A}_u$ by the number of comparison symbols recovers the Čech direction as the associated graded, and the central theorem to prove is the comparison isomorphism between $\mathrm{HP}(\mathbb{A}_u)$, the total cohomology of the model, and Crainic's cyclic theory of the relational groupoid [12] (§4). On this receptacle the curvature is no longer a design question: the Grassmann form $p(dp)^2$ and the Maurer–Cartan form of the descent cocycle are the extreme filtration-components of the curvature $\mathbb{F} = \nabla^2$ of one natural connection on a module over $\mathbb{A}_u$, with Bianchi automatic (§5); the mixed component is the candidate invariant native to relational non-locality, with a predicted home in the non-unit-loop sector of Crainic's decomposition. Graph cohomology enters as the generator and certifier of the cyclic cocycles fed to the characteristic pairing (§7). The remaining foundational questions concern the smooth structure on $\mathbb{A}_u$ — the one place analysis re-enters — convergence of the comparison, and localization at defects (§8).

Remark 0.1 (What changed from v1). The first draft took a Čech–NCG double complex with total differential $D = \check{\delta} \pm d_{\mathcal{V}}$ as the receptacle, and posed the curvature as an open interpolation problem inside it. In this version the receptacle is the cyclic complex of a single algebra with the single differential $b+B$; the bicomplex is demoted to the E_1-page of a filtration; the interpolation problem is solved by a natural construction; and two questions of v1 (the choice of site, refinement invariance as a colimit) dissolve into standard invariance properties of HP. Everything the v1 questions were protecting — that the theory be attached to the system and not to a presentation — is now carried by the algebra itself.

1 Aim, and the template we intend to imitate

The phenomenon we want to make cohomological is the non-locality of quantum interactions: local algebras of observables exist per region, and the physical content that is not localizable in any single region is carried by the *comparisons* between them. The working hypothesis of this proposal is that this non-local content is exactly what Chern–Weil theory is built to measure — provided one can say, discretely and relationally, what plays the role of the bundle, the connection, the curvature, and the invariant polynomial.

Rather than invent the mechanism from nothing, we take as a template the local-to-global architecture of the local RRH theorem [1]. Stripped of its specific coefficients, that architecture is:

(a) an *enlarged flat ambient object* (there: the algebra of formal functions, a Fedosov-type construction [4]), whose flat structure is not a degeneracy but the *realization mechanism* — its flat/ descent-invariant elements recover the honest sheaf $\mathcal{O}(M)$;

(b) a *homological coefficient theory* over the object (there: Hochschild chains of holomorphic differential operators), computed and compared to topological cohomology [3, 20];

(c) an explicit *local-to-global mechanism* with simplex-wise (higher-overlap) compatibilities;

(d) a characteristic map given by *pairing with a universal cocycle* (there: the Feigin–Felder–Shoikhet cocycle τ_{2n} of the Weyl algebra [2]), so that all invariant content is concentrated in the cocycle, none in the differential;

(e) localization of the resulting global class at isolated failures of the mechanism, in the Baum–Bott pattern [18].

The transposition proposed here realizes (a) as a universal algebra $\mathbb{A}_u$ built from the nest itself, (b) as periodic cyclic theory with the single differential $b + B$, (c) as a filtration of the cyclic complex whose associated graded is a Čech complex, (d) as the HP–K-theory pairing with cocycles generated by graphs, and (e) as a relative-cohomology extension problem. Each transfer is nontrivial, and this note tries to state where each one might fail.

Remark 1.1 (Status of everything below). Nothing here is claimed as a theorem except a small number of one-line calculus identities, labelled *Observation*. Statements of intent are labelled *Proposal*; open points are labelled *Question*.

2 The substrate: a nest of algebras with strict descent

Let $\{U_i\}_{i\in I}$ be an abstract cover with overlaps U_{ij} and higher overlaps $U_{i_0\cdots i_p}$. The *relational operator nest* is the data

$$N^{\mathrm{rel}} \;=\; \Big(\{\mathcal{A}_i\}_{i\in I},\ \{\mathcal{A}_{ij}\},\ \rho^i_{ij}:\mathcal{A}_i\to\mathcal{A}_{ij},\ \alpha_{ij}:\mathcal{A}_i\big|_{ij}\xrightarrow{\ \sim\ }\mathcal{A}_j\big|_{ij}\Big), \tag{1}$$

with $\mathcal{A}_i$ unital C^*-algebras of local observables, ρ's the restrictions, and comparison $*$-isomorphisms subject to the *strict* cocycle condition

$$\alpha_{ij}\,\alpha_{jk} \;=\; \alpha_{ik} \qquad \text{on each triple overlap present in the data.} \tag{2}$$

The data descend on the nose to an algebra-sheaf over the comparison site,

$$\mathcal{A}^{\mathrm{rel}}(V) \;=\; \big\{(a_i) \;:\; a_i \in \mathcal{A}_i,\ \alpha_{ij}(a_i|_{ij}) = a_j|_{ij}\big\}, \tag{3}$$

which is *flat*: the comparisons compose strictly, the substrate carries no holonomy, and whatever topology the system has is, at this stage, the combinatorial topology of the cover.

Two precisions. First, the restrictions ρ^i_{ij} are not invertible, so the primitive combinatorial object is a *category* $\mathcal{C}_Q$ of contexts, restrictions and comparisons; the relational groupoid is its invertible part, $\mathcal{R}_Q = \mathrm{Core}(\mathcal{C}_Q)$, in the spirit of sheaves on étale groupoids [12, 13]. Second, the simplexes of the nerve $K = N(\mathcal{R}_Q)$ are *computed* from the data by an incidence rule — $[i_0, \ldots, i_p] \in K$ iff a composable chain of comparisons between these contexts is present in the data — and (2) is read simplex-wise: it is imposed on the 2-simplexes the rule produces, and nowhere else. Only the graphical placement of vertices is free; the incidence structure is an output.

Remark 2.1 (The site question dissolves). The first draft asked which "space" — the nerve, the classifying space $B\mathcal{R}_Q$, or groupoid cohomology — the classes should live on. In the present formulation the invariant is attached to the *algebra* $\mathbb{A}_u$ of §3; the nerve enters only through the presentation of $\mathbb{A}_u$ and the comparison model of §4. Two presentations of the same relational system are expected to give Morita-equivalent (indeed often isomorphic) algebras, and HP is Morita invariant [6, 10]; the residual site-dependence is exactly the dependence of $\mathbb{A}_u$ on the incidence rule, recorded in Question 6.2.

Question 2.2 (Strict descent versus weak descent versus missing simplexes)**.** *Condition* (2) *is a design decision. The alternative — $\alpha_{ij}\alpha_{jk} = \mathrm{Ad}(u_{ijk})\,\alpha_{ik}$ — makes the substrate a stack, with $[u_{ijk}]$ a Dixmier–Douady-type obstruction and "curvature" living in the failure of strictness itself; that is a coherent, arguably more general theory, and this note develops the strict (flat) case. A third branch is distinct from both and is the one adopted here: impose* (2) *simplex-wise on the computed nerve, so that a defect is not a violation of strictness but an* absence of simplexes *— no relation exists there to fail. On the universal algebra this branch is visible structurally (Observation 3.2): a missing* 2*-simplex leaves a free invertible word. Is there a formulation in which the strictness defect of the stack branch and the free holonomy word of the missing-simplex branch appear as two faces of one total obstruction?*

Question 2.3 (Refinement invariance is Morita invariance)**.** *A relational refinement — a morphism of nests compatible with the ρ's and α's — induces a morphism of universal algebras $\mathbb{A}_u(N') \to \mathbb{A}_u(N)$. The expected statement replacing the v1 colimit over covers: a refinement of the cover presenting the same relational system induces a Morita context between the universal algebras, so that the classes of §6 agree under the induced isomorphism of* HP*. What is the correct definition of "presenting the same system," and does every common refinement induce such a context? This is now a question about morphisms of algebras, not about colimits of sites.*

3 The first object: the nest algebra

In the template, the *primary* object is the enlarged flat one; the honest sheaf is *derived* from it. The first object here must therefore be an enlarged, flat, purely algebraic one — with no geometric carrier attached to patches — inside which operator data can be realized, and from which the honest observables are recovered by descent. It should be smaller than the formal enlargement of [1] (no formal coordinates or jets: there is nothing for them to be coordinates *of*), but large enough to contain every local algebra and every comparison as internal structure.

Definition 3.1 (The nest algebra)**.** *The universal algebra of the nest is the quotient of the free product of the $\mathcal{A}_i$ and formal invertible comparison symbols by the nest's own relations,*

$$\mathbb{A}_u \;=\; \Big((\ast_{i\in I}\,\mathcal{A}_i) \ast \mathbb{C}\langle v_{ij}^{\pm 1}\rangle \Big) \Big/ \Big\langle\; v_{ij}\,\rho^i_{ij}(a)\,v_{ij}^{-1} = \rho^j_{ij}\big(\alpha_{ij}(a)\big),\;\; v_{ij}\,v_{jk} = v_{ik} \;\; \textit{on the 2-simplexes of } K \;\Big\rangle. \tag{4}$$

Every $\mathcal{A}_i$ embeds; every comparison becomes inner*; the strict cocycle* (2) *is a defining relation wherever the nerve has 2-simplexes, so flatness is built into the object rather than imposed on a structure over it.*

Realization is then not a stage of the construction but a structural consequence: the universal property of (4) identifies

$$\mathrm{Rep}\big(\mathbb{A}_u,\, H\big) \;\;\cong\;\; \Big\{ \big(\pi_i,\, V_{ij}\big) \;:\; V_{ij}\,\pi_i(a)\,V_{ij}^{*} = \pi_j\big(\alpha_{ij}(a)\big),\;\; V_{ij}V_{jk} = V_{ik} \Big\}, \tag{5}$$

so a compatible family of local representations and implementers *is* a representation π of the first object — realization is co-representable structure, not a step. The recovery statement — the discrete analogue of "flat sections are the honest sheaf" — is

$$H^0\big(\check{C}^\bullet(K;\,\mathbb{A}_u),\;\check{\delta} + \mathrm{Ad}(v)\big) \;\cong\; \mathcal{A}^{\mathrm{rel}}(K), \tag{6}$$

which in degree 0 should be close to a tautology — evidence the framing is right, not that it is empty: the content is in higher degrees, where the same complex reappears as a column of the comparison model of §4.

Observation 3.2 (The holonomy word is free at a defect)**.** *Because $v_{ij}\,v_{jk} = v_{ik}$ is imposed only on the 2-simplexes of the computed nerve, at a defect — a cycle in K bounding no 2-simplex — the word $w = v_{i_n i_1}\cdots v_{i_2 i_3}\,v_{i_1 i_2}$ around the defect loop is unconstrained by the defining relations: an invertible element of $\mathbb{A}_u$ fixed by no relation. The object itself carries the defect as a degree of freedom, and its holonomy is assigned only along a representation π — visible on $\mathbb{A}_u$ before any representation is chosen.* □

Question 3.3 (Completion, and where analysis re-enters)**.** (4) *is a bare $*$-algebra, and this is not a defect to be repaired by a C^*-completion: continuous periodic cyclic cohomology of a C^*-algebra is essentially degenerate (little beyond traces survives), so the receptacle of §4* requires *a smooth structure — a dense Fréchet subalgebra $\mathbb{A}_u^\infty \subset \overline{\mathbb{A}_u}$ stable under holomorphic functional calculus, in the pattern of $C^\infty(M) \subset C(M)$, or else Puschnigg's local cyclic theory [11], which is designed for exactly this defect. Three parts: (i) does $\mathbb{A}_u$ have faithful representations for a general nest, and is a universal completion non-degenerate; (ii) which smooth structure — the choice is the noncommutative shadow of a differentiable structure, and it is arguably the one place genuinely physical input (growth of commutators, locality estimates) enters the formalism; (iii) is $K_0(\mathbb{A}_u^\infty)$ the right home for the geometric data p of §5, i.e. does the inclusion $\mathbb{A}_u^\infty \hookrightarrow \overline{\mathbb{A}_u}$ induce an isomorphism on K_0?*

4 The receptacle: $\mathrm{HP}(\mathbb{A}_u)$, and the Čech comparison

4.1 One algebra, one differential

The intrinsic receptacle of the theory is the periodic cyclic theory of the first object, carried by the single Connes differential:

$$\big(CC_\bullet(\mathbb{A}_u^\infty),\ b+B\big), \qquad \mathrm{HP}_\bullet(\mathbb{A}_u^\infty), \quad \mathrm{HP}^\bullet(\mathbb{A}_u^\infty), \tag{7}$$

with the universal calculus $\Omega_u^\bullet(\mathbb{A}_u^\infty)$, $da = 1\otimes a - a\otimes 1$ [7, 8, 10], as carrier: purely algebraic and functorial, injecting no metric and no curvature. There is no mixed coboundary and no Čech differential in the foundations: because the comparisons are *inner* in $\mathbb{A}_u$, the Čech direction is absorbed into the algebra, and what the first draft treated as a second grading reappears as a filtration.

Definition 4.1 (The v-filtration)**.** *Filter $CC_\bullet(\mathbb{A}_u^\infty)$ by the total number of comparison symbols $v_{ij}^{\pm1}$ occurring in a chain, $F_kCC_\bullet = \{$chains with at most k symbols$\}$. The Hochschild differential b decomposes as $b = b_0 + \check{\delta}$, where b_0 preserves the symbol count and $\check{\delta}$ — the part of b that multiplies a symbol into its neighbour, i.e. shuffles v's past algebra elements — lowers it by one:* the Čech differential is a component of b.

Proposal 4.2 (The bicomplex is the comparison model)**.** *The associated graded of the v-filtration is a Čech complex with cyclic coefficients: the expected identification of the first page of its spectral sequence is*

$$E_1^{p,q} \ \cong\ \check{C}^p\big(K;\ CC_q(\mathcal{A}^{\mathrm{rel}})\big), \tag{8}$$

recovering exactly the Čech–NCG double complex $C^{p,q} = \check{C}^p(K;\mathcal{V}^q)$ of the first draft — with the vertical coefficient no longer a choice *but an output: it is forced to be the sheafified cyclic complex of $\mathcal{A}^{\mathrm{rel}}$. The bicomplex is thereby demoted from receptacle to computational model, in the way Čech–de Rham [17] models de Rham cohomology.*

Proposal 4.3 (The comparison theorem — the central theorem to prove)**.** *Under a convergence hypothesis on the v-filtration (finite-dimensional nerve, or a completed filtration), the spectral sequence of Definition 4.1 converges and induces*

$$\mathrm{HP}_\bullet(\mathbb{A}_u^\infty) \ \cong\ H_{\mathrm{tot}}\Big(\check{C}^p\big(K;\ CC_q(\mathcal{A}^{\mathrm{rel}})\big),\ \check{\delta} \pm (b_0 + B)\Big), \tag{9}$$

identifying the intrinsic single-differential theory with the total cohomology of the Čech–NCG model. This is the discretization statement of the whole proposal: proving (9) is what makes the relational cyclic theory a Čech theory.

Proposal 4.4 (Crainic's theory as groupoid side of the comparison)**.** *The universal algebra maps naturally to the (smooth) convolution algebra of the relational groupoid with coefficients in the sheaf, $\mathbb{A}_u \to C_c^\infty(\mathcal{R}_Q;\,\mathcal{A}^{\mathrm{rel}})$, sending v_{ij} to the indicator of the arrow $j \to i$. Crainic's computation of the cyclic theory of étale groupoid algebras [12, 13] then supplies the groupoid-side evaluation of the right-hand side of (9): the proposal is a Chern–Weil* presentation *of that theory, not a rival to it. Crainic's localization decomposes the cyclic theory over the loops of the groupoid; the unit-loop sector should receive the geometric (Grassmann) classes of §5, and the non-unit sectors are inhabited exactly by the free holonomy words of Observation 3.2 — see Question 5.4.*

Question 4.5 (Convergence, and the fineness hypothesis retargeted)**.** *Two analytic hypotheses now sit inside the comparison theorem rather than inside the theory. (i) Convergence of the v-filtration spectral sequence: for an infinite or infinite-dimensional nerve, does the filtration have to be completed, and does* (9) *survive completion? (ii) Acyclicity of the coefficient: for the E_1-page to compute the derived functor — the v1 "fineness" question — do partitions of unity have a relational substitute (idempotent splittings, quasi-central approximate units)? Both are hypotheses of Proposal 4.3; neither is needed to* define *the theory, which is* (7)*.*

5 The curvature

Two demands were placed on "curvature" in the first draft, and they pulled in opposite directions: it must carry the *geometric content of the algebra* — modules, projections, K-theory — (G); and it must be compatible with the *flat* comparison structure, because flatness is the realization mechanism (F). The Grassmann form $F = p(dp)^2$ [6, 7] satisfies (G) and is blind to the relational direction; the Maurer–Cartan data $a_{ij} = dV_{ij}$, $F^{\mathrm{rel}}_{ijk} = a_{ij}a_{jk}$ of the descent cocycle satisfies (F) and is blind to K_0. On $\mathbb{A}_u$ the tension dissolves, because both live in one calculus, in different filtration degrees; the interpolating object is no longer a proposal but a construction.

Definition 5.1 (The natural connection and its curvature)**.** *Let $\omega := \sum_{[ij]\in K_1} dv_{ij}\, v_{ij}^{-1} \in \Omega^1_u(\mathbb{A}_u^\infty)$ be the comparison potential (the sum over the* 1*-simplexes of the nerve; finite cover assumed), and let $p \in M_N(\mathbb{A}_u^\infty)$ be an idempotent — the geometric datum, a sector. On the module $E = p\,(\mathbb{A}_u^\infty)^N$ set*

$$\nabla \;:=\; p \circ \big(d + \omega\big) \;:\; E \longrightarrow \Omega^1_u \otimes_{\mathbb{A}_u^\infty} E, \qquad \mathbb{F} \;:=\; \nabla^2. \tag{10}$$

Expanding, the curvature has three components, graded by the v-filtration of Definition 4.1:

$$\mathbb{F} \;=\; \underbrace{p\,(dp)^2}_{v\text{-count } 0} \;+\; \underbrace{p\big(dp\cdot\omega + \omega\cdot dp\big)p + \cdots}_{v\text{-count } 1} \;+\; \underbrace{p\,(d\omega + \omega^2)\,p}_{v\text{-count } 2}, \tag{11}$$

*whose extreme components reproduce, under the comparison of §4, the two candidates of the first draft: the v-count-*0 *part is the Grassmann curvature, and the v-count-*2 *part maps to the Čech* 2*-cochain $a_{ij}a_{jk}$ under any representation* (5)*.*

Observation 5.2. $[\nabla, \mathbb{F}] = [\nabla, \nabla^2] = 0$ *identically; there is one Bianchi identity for the one curvature. Component wise it contains the two identities of the first draft ($\check{\delta}F^{\mathrm{rel}} = 0$ from Leibniz on the strictly preserved relations; $p(dp)p = 0$ from $p^2 = p$) as its extreme filtration degrees, and the mixed identity — formerly an open requirement — as the middle one. Likewise the degenerations are built in: p supported in a single patch and descent-trivial $\Rightarrow$ $\mathbb{F}$ reduces to the Grassmann class of $[p]$; $p = 1 \Rightarrow \mathbb{F} = d\omega + \omega^2$, whose Čech shadow is F^{rel}.* ☐

Question 5.3 (Consistency of the extreme components)**.** *Under what hypotheses on the realization do the extreme components compute the same class where both are defined — i.e. when does $\tau\big(p_i\, dV_{ij}\, dV_{jk}\big)$ agree with the Connes–Chern character of $[p]$ built from $p(dp)^2$? This is the internal consistency check of the design, now a computation inside one calculus, and the first thing to try to prove or refute.*

Question 5.4 (Is the mixed component the new invariant?)**.** *The v-count-*2 *part is classical descent, the v-count-*0 *part is classical NCG; only the mixed component — lowest order $p_j - v_{ij}\, p_i\, v_{ij}^{-1}$, the incompatibility of the sector with the comparisons — is native to the relational setting. Is* *there a nonzero class of $\mathbb{F}$ whose extreme components vanish — pure relational non-locality? Proposal 4.4 supplies a predicted home: the non-unit-loop sectors of Crainic's decomposition, where the free holonomy words of Observation 3.2 live. Exhibiting one nonzero class there, and identifying what it pairs with, would be the first genuinely new number of the theory.*

6 The characteristic map as a pairing

In LRRH the characteristic map is cup product with a fixed universal cocycle, all geometry in the cocycle, globalization by the simplicial cup product [2, 1]. Here the same role is played by the HP–K-theory pairing on the single algebra: for τ a cyclic cocycle on $\mathbb{A}_u^\infty$ ($(b+B)\tau = 0$),

$$\chi_\tau(\mathbb{F}) \;\coloneqq\; \tau\big(\mathbb{F}^m\big) \;\in\; \mathbb{C}, \tag{12}$$

the Chern–Weil evaluation of the one curvature against the one cocycle. Under the comparison of §4 this *becomes* the simplicial cup-product formula of the first draft — e.g. in lowest degree $c_{ijk} = \tau\big(p_i\, dV_{ij}\, dV_{jk}\big)$, closed by Observation 5.2 together with $(b+B)\tau = 0$, defining a class in $\check{H}^{2m}(K;\mathbb{C})$ — so the discrete cup product is the shadow of the cyclic pairing, not an independent mechanism. Modularity is manifest: substrate, flat structure, and Bianchi are fixed once; τ alone selects the invariant.

Because the class is defined on $\mathbb{A}_u^\infty$ *before* any representation, extraction of numbers is a restriction: a realization π as in (5) supplies the analytic content (a trace, or the Fredholm module defining τ), and different invariants on the same substrate are different restrictions (π, τ) of one universal class.

Question 6.1 (Which cocycles?)**.** *In LRRH the cocycle is canonical because every local model is the* same *algebra and τ_{2n} is its essentially unique normalized cocycle [2]. Relationally the $\mathcal{A}_i$ carry no canonical identification with a model. Candidates for $\tau \in \mathrm{HP}^\bullet(\mathbb{A}_u^\infty)$: (i) the Connes–Chern character of a finitely summable Fredholm module [6] — canonical once the module is chosen, but the choice re-imports geometry; (ii) a trace alone, raised in degree by powers of $\mathbb{F}$ — minimal, possibly too poor; (iii) a universal cocycle on a local model plus the comparisons — the closest transcription, but requiring a local-triviality hypothesis the relational setting was designed to avoid. Deciding among (i)–(iii), or subsuming them, remains in our view the second-hardest open point (after Proposal 4.3). §7 proposes where the answer should come from.*

Question 6.2 (Presentation independence)**.** *Most of the v1 representation-independence question is now absorbed by standard properties of* HP*: Morita invariance and invariance under inner perturbations [6, 10] handle Morita-equivalent realizations and the ambiguity of the implementers V_{ij} (fixed by α_{ij} only up to the commutant's unitary group). What remains is sharper and new: (i) independence of the incidence rule producing the nerve, i.e. of the presentation of $\mathbb{A}_u$ (Remark 2.1); and (ii) independence of the choice of smooth structure $\mathbb{A}_u^\infty$ within a natural class (Question 3.3(ii)) — or, if the classes genuinely depend on it, the physical interpretation of that dependence.*

7 Graph cohomology: how should it be used?

Graph complexes know nothing about the substrate; their competence is the space of *universal operations* — universal cocycles, formality morphisms, and the $\mathfrak{grt}_1$-action on them [16, 14, 15];

the FFS cocycle is itself a sum over graphs [2]. Their role here is to generate and certify the inputs τ of the pairing (12).

Proposal 7.1 (Graphs generate the characteristic cocycles)**.** *Seek a morphism of complexes*

$$\Phi\ :\ \big(\mathsf{GC}',\ \delta_{\text{graph}}\big)\ \longrightarrow\ \big(CC^{\bullet}(\mathbb{A}_u^{\infty}),\ b+B\big),\qquad \Gamma\ \longmapsto\ \tau_{\Gamma},\tag{13}$$

assigning to each admissible graph a cyclic cochain (vertices $\mapsto$ algebra insertions, edges $\mapsto$ universal-calculus contractions), such that:

(i) graph cocycles map to cyclic cocycles, so every closed $\tau=\sum_{\Gamma} w_{\Gamma}\,\tau_{\Gamma}$ is an admissible input to (12) *— graphs* supply *the answers to Question 6.1;*

(ii) exact graphs map to $(b+B)$-coboundaries, so the class $\chi_{\tau}[\mathbb{F}]$ is independent of the graphical presentation of τ, and the symmetry group of presentations ($\mathfrak{grt}_1$-type, as in [16]) acts trivially on it;

(iii) the graph-complex products are carried to the product of the cyclic theory (hence, under the comparison of §4, to the simplicial cup product on the model), so higher classes $\chi_{\tau}(\mathbb{F}^m)$ are computed graph-by-graph.

If (i)–(iii) hold, graph cohomology is precisely the cohomology of the characteristic map's inputs: it classifies what can be paired with, and certifies that the resulting class does not depend on how it was written.

Question 7.2. *Does* (13) *exist over $\mathbb{A}_u^{\infty}$ with purely combinatorial weights w_{Γ} — no configuration-space integrals, whose analytic meaning is unavailable here? A failure mode worth anticipating: the chain-map property may force relations among the w_{Γ} that only integral weights satisfy. (Which variant of the graph complex — GC_2, oriented, ribbon — is a real but downstream choice; a couple of computations with* (13) *in low degree should decide it.)*

8 Target and localization

Question 8.1 (What is the theorem about?)**.** *A Chern–Weil mechanism must compute something; in LRRH, a trace formula. The natural discrete target is the index pairing*

$$\big\langle\tau,\,[p]\big\rangle\in\mathbb{Z},\qquad [p]\in K_0(\mathbb{A}_u^{\infty}),\tag{14}$$

integral when τ is the Connes–Chern character of a finitely summable Fredholm module [6], in the pattern of [19]. We record this as the intended target, not a result. Is there a statement of the form "for every finitely summable relational realization, $\chi_{\tau}(\mathbb{F})$ is an index" — a discrete RRH?

Question 8.2 (Localization at an augmentation vertex)**.** *The template localizes: obstructed at isolated points, the global class concentrates there as residues [18, 1]. Relationally a defect is a place where simplexes are* absent *(Question 2.2), and on the algebra it is a free holonomy word w (Observation 3.2); localization is an* extension *problem. Adjoin to $\mathcal{C}_Q$ a freely added terminal object d — no algebra, no patch, no operator data — i.e. pass on nerves to the simplicial cone $C_dK=d*K$. A cochain c built from the physical data on K may or may not extend to $\tilde{c}$ on C_dK with $\tilde{c}|_K=c$ and $\tilde{c}$ closed: the vertex adds a constraint, asking whether the pairwise comparisons of K participate in one global structure. Since the cone is contractible, the content is relative,*

$$\check{H}^{\,p+1}\big(C_dK,\,K\big)\ \cong\ \tilde{\check{H}}^{\,p}(K),$$

so the class concentrates at the defect as a relative class computable from boundary data alone — the Baum–Bott pattern, with the cone playing the star. Does the class $\chi_\tau(\mathbb{F})$ *localize in this sense, and is the resulting obstruction a function of the represented holonomy* $\pi(w)$ *of the free word — equivalently, is the defect residue computed in the non-unit sector of Proposal 4.4? Stated purely about descent data, this remains the part of the template where the transfer from [1] looks most nearly verbatim — perhaps the right place to attempt a first complete proof.*

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